

Kimi-K3-Neuron-IQ1S-GGUF — independent bench report

Bench date: 2026-08-08 · **Benchd by:** NetSphere estate (Killy), automated by Claude

Hardware: 4× RTX PRO 6000 Blackwell Workstation Edition (sm_120, 96 GB each, 384 GB total), 52 vCPU / 639 GiB RAM, NVMe (measured 13.1 GB/s)

Engine: llama.cpp custom sm120/cuda13.3 build, single stream (--parallel 1)

Launch flags (identical to our official-IQ1_S baselines): ctx 262144 · ub 2048 · flash-attn on · temp 1.0 / top-p 0.95 / top-k 0 / min-p 0 · reasoning_effort=max · --reasoning-budget 16384 · --no-cache-prompt -sps 0 (K3 multi-turn KDA-state correctness flags)

Artifact verification

- 9 shards, **307.49 GiB**, byte-sizes match the repo listing.
- GGUF header: arch kimi-k3, 93 blocks, 896 experts / 16 active / 4 shared — **full expert structure retained**, as the card claims.
- expert_feed_forward_length **1536 vs 3072 official** — the pruning is exactly **50% of every expert's FFN channels**; dense FFN (33792) and embeddings (7168) untouched.
- **No nextn/MTP tensors — but the official unsloth GGUF has none either** (scanned all 14 shards). Nothing was lost to pruning on that front; MTP-based speculative decode was never available to K3 GGUFs on this engine.
- Load-time warning special_eos_id is not in special_eog_ids (tokenizer config): **benign in practice** — generation converges and stops correctly across 100+ eval items.
- Loads healthy in **21–30 s** from page cache and goes **fully GPU-resident (~355 GB incl. KV/buffers)** — the first K3 quant that fits this estate's VRAM. The 555 GB official IQ1_S must spill ~200 GB of experts to CPU/mmap on the same hardware.

Speed (same harness, same flags, same machine as baselines)

metric	official UD-IQ1_S (555 GB, CPU-spill)	Neuron (330 GB, GPU-resident)	×
deep prefill @ 4K / 16K / 32K prompt	123 / 150 / 158 tok/s	581 / 665 / 641	4.1–4.7×
deep decode @ 4K / 16K / 32K	12.29 / 12.30 / 12.19	20.21 / 20.01 / 19.79	1.6×
short-prompt decode	~11.5	20.3	1.8×
100K-prompt prefill (deepctx replay)	84.6 (q2kxl datum)	423.6–424.2	~5×

Config matrix (pre-registered, one variable at a time, ≥2 reps): flash-attn on/off/auto, ubatch 512/2048/4096, threads 8/52, sampler 2×2 (top_k × min_p). **Incumbent config optimal on every axis.** fa off cannot even allocate at 262K ctx; ub2048 = ub4096 > ub512 on prefill (664 vs 399); threads and sampling are speed-neutral. Decode is kernel-latency-bound (tiny 1536-wide experts), insensitive to every lever tried — the gain over official is residency, and it saturates there.

Quality

Full v2 suite (80 scored items, seed 7, ctx-tokens 100K, deepctx 15 / agentic 15 / grounded 30 / math-hard 20):

model	agent	deepctx	grounded	math-hard	OVERALL
GLM TR3 (estate resident, 90 tok/s)	15/15	7/15	30/30	19/20	71/80
k3-neuron (18 tok/s)	13/15	15/15	29/30	11/20	68/80

Overall statistically indistinguishable from the resident (McNemar $p = 0.65$) — but the categories are violently opposed, both at **$p = 0.0078$** :

- **deepctx 15/15, perfect at every needle depth (5/5 at $d=0.1/0.5/0.9$)** vs TR3's 7/15. On the hardest-5 replay (120K-token prompts): **4/5, including a PASS on contradiction-009 — an item with a 0-7/22 field record that every previously-benched model failed.**
- **math-hard 11/20** vs 19/20. Caveat in your favor: our server caps reasoning at 16,384 tokens while the TR3 run thought up to 32K, so part of this gap is budget asymmetry, not capability.

Same-model control (narrow-15 vs our unpruned 803 GB Q2_K_XL, identical server flags and budget): **9/15 vs 15/15** ($p = 0.0312$). Failure anatomy on math-hard (1/5 vs 5/5):

- **3 × non-convergence** — finish=length at exactly the reasoning budget; the model reasons without landing where the unpruned quant closes in the same budget.
- **1 × digit-slip** — converged but 53401/24266 vs correct 53401/24366.
- Agentic (3/5): two failures produced **empty final answers** after completed tool loops — protocol degradation, not wrong values.
- Grounded retrieval: clean (5/5 narrow, 29/30 full).

The budget-vs-pruning attribution (follow-up run)

The 16K reasoning cap was our server's config, not yours, so we re-ran the nine full-v2 math failures at a **32,768 reasoning budget**, then fed whatever still failed to the **official unpruned IQ1_S at the same budget** — isolating budget from pruning from bitwidth.

- **6 of 9 rescued at 32K** (converged at 10.1–18.6K thinking tokens — past the old cap). Neuron's honest math-hard at 32K is **17/20**, and its full-v2-equivalent rises to **~74/80 — above the estate resident's 71/80**, with deep-context doing the lifting.
- **The 3 stubborn items went to the official IQ1_S at 32K, which solved 2 — fast.** Same bitwidth, no pruning: on those items the pruning is the culprit, and the mechanism is **reasoning efficiency** — 2–5× the thinking tokens to close an argument — not a capability ceiling.
- 13_recur fails the official IQ1_S too (fast but wrong) while our 803 GB Q2_K_XL solves it — a **bitwidth casualty**, not yours.

item	Neuron @ 32K	official IQ1_S @ 32K
math_hard_03_det	FAIL — burned 32,768, no answer	PASS — 6,551 tokens

item	Neuron @ 32K	official IQ1_S @ 32K
math_hard_12_order	FAIL — 25,083 tokens, wrong (15 vs 58)	PASS — 3,518 tokens
math_hard_13_recur	FAIL — burned 32,768, no answer	FAIL — 4,727 tokens, wrong (3 vs 116)

Verdict

The pruning method preserved attention/retrieval **perfectly** — consistent with the design (channels cut inside expert FFNs only; attention kept Q8) — and the cost is concentrated in **deep serial reasoning**, expressed mostly as non-convergence rather than wrong answers, plus occasional last-digit numeric slips. Your card's 73.7% top-1 agreement disclosure is consistent with what we measured.

As a long-context reading/extraction body this is exceptional per-GB — it beat our 400K-context resident on deep context while being 44% smaller than your own size class. As a math/agent brain, the unpruned Q2_K_XL remains clearly better at equal budget. If the method allows a selective variant (e.g., prune shallower on layers that host reasoning-heavy experts, or 60/40 instead of 50/50), the deepctx result says the retrieval side has margin to spare.

Not tested: your WikiText-2 PPL protocol (no perplexity binary in our engine image) and HumanEval (different harness). Raw artifacts available: config matrix jsonl, ladder/deep JSONs, full netterm-eval run dirs with per-item transcripts.